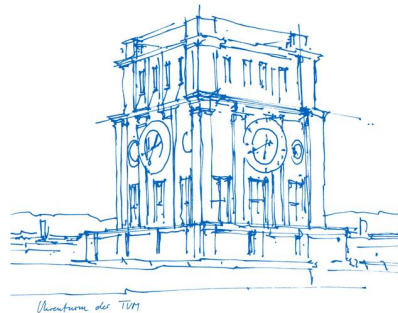


PoreAccess-CO₂: From Random Packing to an Interpretable Transport Closure

Structure-resolved diffusion and adsorption in DEM-generated zeolite beds

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2026-09-11

Good morning. This project is called PoreAccess-CO₂. It examines how the random arrangement of particles changes CO₂ transport through a packed zeolite bed. The main aim is to retain information about the pore structure while developing a model that is much faster than a complete pore-network adsorption simulation. I will first introduce the physical problem, then explain the pore-network and one-dimensional models. Finally, I will show how we connect them through an accessibility-based transport closure and what the first twenty packing realizations tell us.



CO₂ adsorption matters and packing structure is a missing scale



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Why CO₂ capturing matters?

- Food systems contribute about **one-third** of global anthropogenic greenhouse-gas emissions.
- Food and bioprocessing include recoverable biogenic CO₂ streams, for example from fermentation.
- Regenerable sorbents offer a compact capture and purification route.

What is already established?

- CO₂ equilibrium and uptake kinetics on zeolite 13X are widely studied.
- Fixed-bed models normally use bulk porosity and a fitted effective transport coefficient.
- Random inlet-connected paths remain hidden inside that coefficient.

Novelty: connect random particle geometry to a fast bed-scale uptake model through a physically calculated pore-accessibility descriptor.

CO₂ adsorption matters and packing structure is a missing scale

CO₂ capture and separation matter across many sectors. Food systems contribute roughly one third of anthropogenic greenhouse-gas emissions, and food and bioprocessing also generate concentrated biogenic CO₂ streams, for example during fermentation. Adsorption offers one possible route for capturing or purifying such streams.

The equilibrium behaviour and uptake kinetics of CO₂ on zeolite 13X have already received extensive attention. At bed scale, however, models usually represent a random packing only through its average porosity and an effective transport coefficient. That treatment hides whether the void space forms several open paths from the inlet or only a few restricted paths. Our novelty lies in connecting the random particle geometry to a fast bed-scale uptake model through a physically calculated accessibility descriptor.

CO₂ adsorption matters and packing structure is a missing scale

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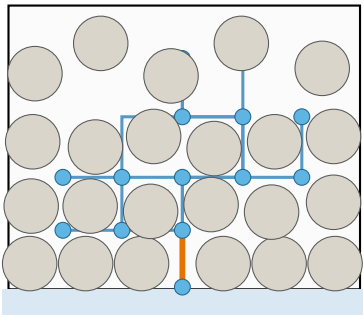
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Novelty: connect random particle geometry to a fast bed-scale uptake model through a physically calculated pore-accessibility descriptor.

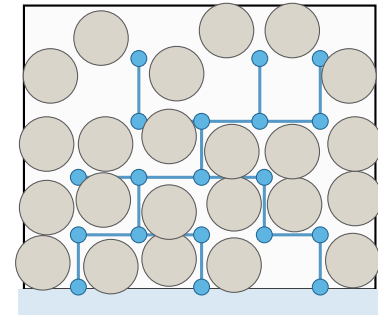
Same porosity can hide different inlet-connected pathways

Can a pore-accessibility measure predict bed-scale transient transport better than porosity alone?

Packing A: inlet bottleneck



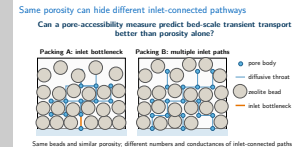
Packing B: multiple inlet paths



- pore body
- diffusive throat
- zeolite bead
- inlet bottleneck

Same beads and similar porosity; different numbers and conductances of inlet-connected paths.

Same porosity can hide different inlet-connected pathways



This schematic introduces the central problem. The two beds contain the same monodisperse beads and can have almost the same total porosity. Their inlet-connected paths can nevertheless differ strongly. Packing A illustrates a network reached through one restrictive inlet bottleneck. Packing B illustrates several paths that connect the reservoir to the bed interior.

The blue lines show only simplified inlet-connected paths, not every throat in a real extracted network. The sketches are therefore hypotheses rather than simulation results. They motivate our research question: can a quantitative pore-accessibility measure predict transient bed-scale transport better than bulk porosity alone?

Why zeolite 13X and what does the study isolate?

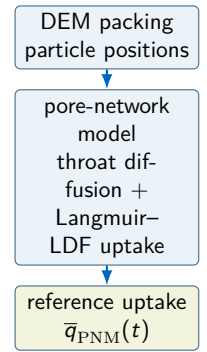
Why zeolite 13X?

- established benchmark sorbent for CO₂
- strong equilibrium uptake at low partial pressure
- commercial millimetre-scale beads form random beds

Controlled model problem

- dry, isothermal, stagnant gas
- CO₂ enters from a reservoir below the bed
- pore-network model resolves throat diffusion and bead adsorption
- no imposed flow; no CFD

How strongly does random void connectivity change access to the adsorbent?



Why zeolite 13X and what does the study isolate?

We use zeolite 13X because it is a well-established benchmark sorbent for CO₂. It has strong equilibrium uptake at low partial pressure and is available as millimetre-scale beads that naturally form random beds.

The study deliberately isolates one mechanism. The gas is dry, isothermal and stagnant. CO₂ enters from a reservoir below the bed. We do not impose a gas flow and we do not solve CFD. We do, however, model the pore space explicitly. Starting from the DEM particle positions, the pore-network model resolves diffusion through the throats and couples it to Langmuir equilibrium and finite-rate adsorption in the beads. Its principal reference output is the mean network uptake curve, $\bar{q}_{\text{PNM}}(t)$.

Twenty beds isolate variability caused by random arrangement



Quantity	Fixed value
beads per bed	1850
bead diameter	1.6 mm
tube diameter	16 mm
D_{tube}/d_p	10
temperature	298.15 K
inlet c_{CO_2}	6.05 mol m ⁻³
simulation time	5000 s

Only the DEM insertion seed changed

- Delaunay–Voronoi networks were built directly from particle coordinates
- all networks were connected and spanning
- quasi-uniform Sobol sampling allocated pore volumes without a voxel grid
- transient and 1D models passed mass-balance checks

The ensemble tests arrangement effects under otherwise identical conditions.

2026-09-11

Twenty beds isolate variability caused by random arrangement

We generated twenty independent beds. Every listed physical and numerical condition remains fixed, including the number and diameter of the beads, tube diameter, temperature, inlet concentration and simulation duration. Only the DEM insertion seed changes. Differences between the cases can therefore be attributed to the random arrangement rather than to changed operating conditions.

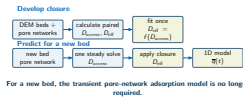
We construct a Delaunay–Voronoi pore network directly from the particle coordinates. Sobol sampling means a deterministic, low-discrepancy set of quasi-uniform points. We use these points only to allocate the analytically calculated void volume among pore bodies. They do not create the network connectivity, and no voxel grid is required. All twenty networks span the bed and pass the mass-balance checks.

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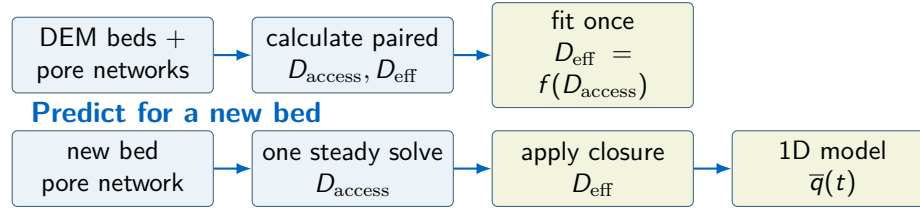
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The ensemble tests arrangement effects under otherwise identical conditions.



Development and later use are two different workflows

Develop closure



For a new bed, the transient pore-network adsorption model is no longer required.

This slide separates development from later prediction. During development, we run the full transient pore-network adsorption model for each training bed. From its uptake curve, we fit one effective diffusivity, D_{eff} . Independently, one steady network solve gives the accessibility diffusivity, D_{access} . The paired values across the beds allow us to fit the closure $D_{\text{eff}} = f(D_{\text{access}})$. For a new packing, the lower workflow applies. We still extract its pore network, but we perform only the inexpensive steady accessibility solve. The closure predicts D_{eff} , and the 1D model then predicts the complete uptake curve. The expensive transient pore-network adsorption calculation belongs to development and validation, not to routine use.

Pore-network geometry and throat diffusion

Pore-network geometry and throat diffusion	
Network construction	Diffusion through throat $i-j$
- four bead centres define a Delaunay tetrahedron; its circumcentre is a pore body - tetrahedra sharing a triangular face are joined by a throat - topology comes from particle coordinates, not voxels	$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$ $L_{ij} = \ \mathbf{x}_i - \mathbf{x}_j\$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area $A_{ij} = \pi r_{ij}^2$, where r_{ij} is the clearance of the shared Delaunay face. The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{s}^{-1}$.

Network construction

- four bead centres define a Delaunay tetrahedron; its circumcentre is a pore body
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels

A low-discrepancy Sobol sample of 2^{22} points allocates the analytically calculated void volume to pore bodies; it does not define connectivity.

Diffusion through throat $i-j$

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where r_{ij} is the clearance of the shared Delaunay face.

The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{s}^{-1}$.

Four neighbouring bead centres define a Delaunay tetrahedron, and its circumcentre becomes a pore body. Two tetrahedra sharing a triangular face create two neighbouring pores connected by a throat. The indices i and j therefore refer to adjacent pore bodies.

Diffusion through that throat follows $G_{ij} = D_m A_{ij} / L_{ij}$. Here D_m has units of square metres per second, A_{ij} has units of square metres and L_{ij} has units of metres. Thus, G_{ij} has units of cubic metres per second. Multiplying it by the concentration difference gives the molar flow rate through the throat. The actual opening is irregular. We approximate it by a circular area $A_{ij} = \pi r_{ij}^2$. This equivalent circular area is an explicit geometric assumption of the model.

Langmuir equilibrium and bead uptake

Langmuir equilibrium and bead uptake	
Local equilibrium loading	Meaning of the rate equation
$p_k = c_k RT, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}$	q_k is the current loading of bead k k_{LDF} controls how quickly q_k approaches q_k^* - gas removed from adjacent pores is added to the bead
Finite-rate uptake	Pore-network outputs
$\frac{dq_k}{dt} = k_{\text{LDF}} (q_k^* - q_k)$	- pore concentrations $c_i(t)$ - particle loadings $q_k(t)$ - mean uptake $\bar{q}_{\text{PNM}}(t)$
q_s is the saturation loading, b is the Langmuir affinity, p_k is the local CO_2 partial pressure, and q_k^* is the corresponding equilibrium loading. The product $b p_k$ is dimensionless. The coupled diffusion-adsorption calculation conserves the total amount of CO_2 .	

Langmuir equilibrium and bead uptake

Local equilibrium loading

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Finite-rate uptake

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Meaning of the rate equation

- q_k is the current loading of bead k
- k_{LDF} controls how quickly q_k approaches q_k^*
- gas removed from adjacent pores is added to the bead

Pore-network outputs

- pore concentrations $c_i(t)$
- particle loadings $q_k(t)$
- mean uptake $\bar{q}_{\text{PNM}}(t)$

Diffusion supplies a local gas concentration c_k around bead k . We convert it to partial pressure through $p_k = c_k RT$. The Langmuir equation then gives the equilibrium loading q_k^* . The parameter q_s is the saturation loading and b is the Langmuir affinity. Because b has units of inverse pressure, the product $b p_k$ is dimensionless.

The actual particle loading q_k does not jump immediately to equilibrium. The linear-driving-force equation describes how it approaches q_k^* , with k_{LDF} setting the uptake rate. Gas removed from the pores is added to the particles, so the coupled calculation conserves CO_2 . The detailed outputs include every pore concentration $c_i(t)$ and every bead loading $q_k(t)$. Averaging the bead inventory gives the reference uptake curve $\bar{q}_{\text{PNM}}(t)$.

The homogeneous 1D uptake model

Axial diffusion and adsorption

$$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t},$$

$$\frac{\partial q}{\partial t} = k_{\text{LDF}} [q^*(c) - q].$$

The bed is divided into 100 axial cells. A trial diffusivity D gives concentration $c(z, t)$ and loading $q(z, t)$.

For every trial D , the 1D solver produces a complete mean uptake curve.

Quantity compared with the network

$$\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$$

The 1D model uses the same bed height, porosity, inlet condition, Langmuir parameters, LDF coefficient and initial state as the pore network. Only D changes during fitting.

2026-09-11

The homogeneous 1D uptake model

The reduced model replaces the irregular pore network by one axial coordinate. It solves coupled diffusion and adsorption in one hundred cells. The local outputs are the gas concentration $c(z, t)$ and loading $q(z, t)$. Their spatial average gives $\bar{q}_{1D}(t; D)$, the mean uptake predicted by the 1D model.

The bed height, porosity, inlet condition, Langmuir parameters, LDF coefficient and initial state remain identical to those in the pore-network calculation. Only the diffusivity D changes. For any selected value of D , the solver must integrate the complete transient problem and return a complete uptake curve. This is why we call it a trial diffusivity at this stage. We do not yet know which value best represents the network.

The homogeneous 1D uptake model

Axial diffusion and adsorption $\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t}$ $\frac{\partial q}{\partial t} = k_{\text{LDF}} [q^*(c) - q].$ The bed is divided into 100 axial cells. A trial diffusivity D gives concentration $c(z, t)$ and loading $q(z, t)$.	Quantity compared with the network $\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$ The 1D model uses the same bed height, porosity, inlet condition, Langmuir parameters, LDF coefficient and initial state as the pore network. Only D changes during fitting. For every trial D, the 1D solver produces a complete mean uptake curve.
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D_{access} and D_{eff} answer different questions

D_{access} : geometric accessibility

- fix concentration at inlet and interior plane
- solve one steady, linear diffusion problem
- no adsorption and no time integration

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

G_{access} is total steady flux per concentration difference. H is bed height; A_{tube} is tube area.

D_{eff} : transient reduced-model coefficient

- full transient network gives $\bar{q}_{\text{PNM}}(t)$
- repeated 1D solves give $\bar{q}_{1\text{D}}(t; D)$

Define the curve residual

$$e_m(D) = \bar{q}_{1\text{D}}(t_m; D) - \bar{q}_{\text{PNM}}(t_m).$$

$$\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{\text{PNM}}(t_f)},$$

$$D_{\text{eff}} = \arg \min_D \text{NRMSE}(D).$$

The trapezoidal weights w_m account for unequal time spacing and satisfy $\sum_m w_m = 1$.

D_{access} is cheap because it avoids adsorption dynamics and optimization.

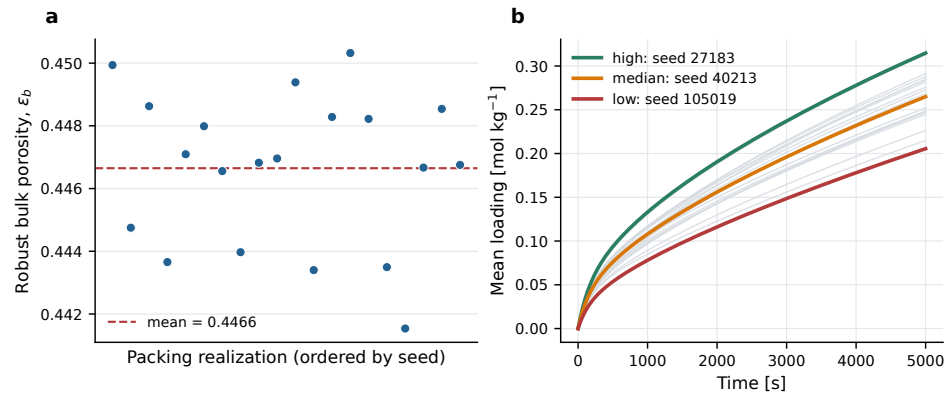
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These two quantities answer different questions. To calculate D_{access} , we prescribe concentrations at the inlet and at an interior plane and solve one steady linear diffusion problem. Steady here means that we directly solve the time-independent balance. We do not run a transient simulation and wait for physical steady state. The result is the total conductance G_{access} between the two boundaries. Multiplying it by bed height and dividing by tube area expresses it in diffusivity units.

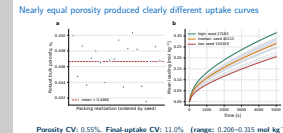
D_{eff} is different. It is the coefficient required by the reduced 1D transient model. The network first supplies $\bar{q}_{\text{PNM}}(t)$. We choose a trial D , solve the 1D transient model and obtain $\bar{q}_{1\text{D}}(t; D)$. The optimizer then changes D and repeats the 1D solve. At every saved time, $e_m(D)$ is the difference between the 1D and network uptake. NRMSE combines these errors over the full curve. Squaring prevents positive and negative errors from cancelling, the square root restores loading units, and division by the final network loading makes the result dimensionless. The time weights account for unequal intervals. The value that minimizes NRMSE is called D_{eff} . This optimization explains why obtaining D_{eff} is more expensive than calculating D_{access} .

Nearly equal porosity produced clearly different uptake curves



Porosity CV: 0.55%. Final-uptake CV: 11.0% (range: 0.206–0.315 mol kg⁻¹).

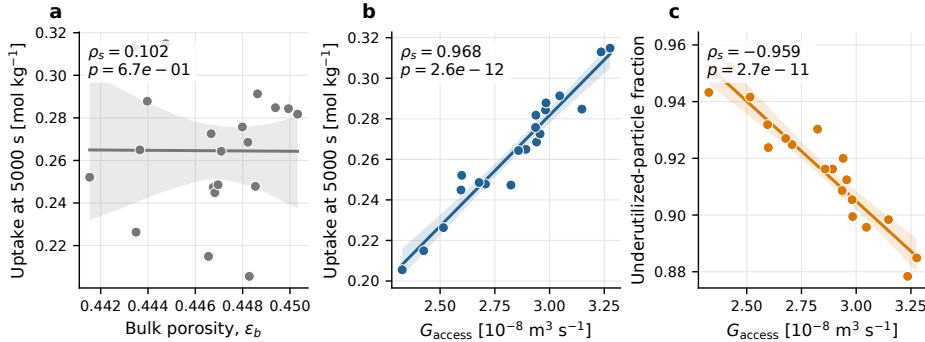
Nearly equal porosity produced clearly different uptake curves



Panel a shows that the twenty beds have closely matched bulk porosity. The mean is approximately 0.44665. The sample standard deviation divided by this mean gives a coefficient of variation of only 0.55 percent.

Panel b shows a different picture for adsorption. The grey lines represent all twenty network uptake curves. The highlighted curves correspond to beds with low, median and high accessibility. At 5000 seconds, the mean loading ranges from about 0.206 to 0.315 moles per kilogram, producing a coefficient of variation of 11 percent. Thus, closely matched void volume does not produce closely matched transient utilization. Something about the arrangement and connectivity of that void space must explain the difference.

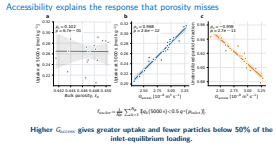
Accessibility explains the response that porosity misses



$$f_{\text{under}} = \frac{1}{N_p} \sum_{k=1}^{N_p} \mathbb{I}[q_k(5000 \text{ s}) < 0.5 q^*(p_{\text{inlet}})].$$

Higher G_{access} gives greater uptake and fewer particles below 50% of the inlet-equilibrium loading.

Accessibility explains the response that porosity misses



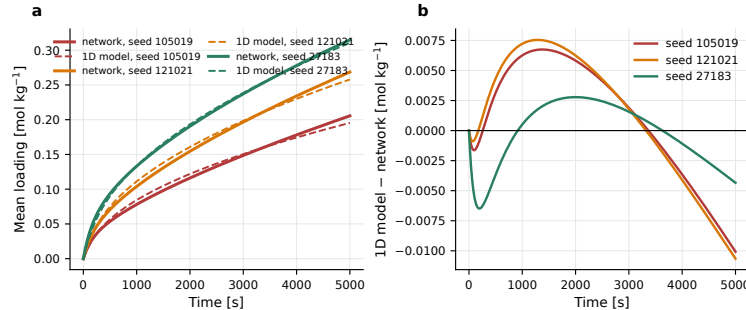
Panel a confirms that bulk porosity has almost no monotonic relationship with final uptake. Panel b shows a strong positive relationship between accessibility conductance and uptake. The Spearman coefficient is 0.968, meaning that beds ranked as more accessible generally also rank higher in final uptake.

Panel c examines local particle utilization. We define a particle as underutilized when its loading after 5000 seconds remains below half of the equilibrium loading corresponding to the inlet pressure. The symbol $\mathbb{I}$ is an indicator function. It equals one when this condition is true and zero otherwise. Averaging the indicator over all particles gives the underutilized fraction. Higher G_{access} corresponds to fewer underutilized particles. The descriptor therefore relates not only to the mean uptake but also to how effectively the bed interior is reached.

For each bed, D_{eff} is fitted to the complete uptake curve

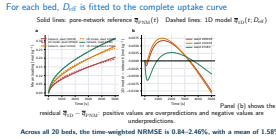
For each bed, D_{eff} is fitted to the complete uptake curve

Solid lines: pore-network reference $\bar{q}_{\text{PNM}}(t)$ Dashed lines: 1D model $\bar{q}_{1\text{D}}(t; D_{\text{eff}})$



Panel (b) shows the residual $\bar{q}_{1\text{D}} - \bar{q}_{\text{PNM}}$: positive values are overpredictions and negative values are underpredictions.

Across all 20 beds, the time-weighted NRMSE is 0.84–2.46%, with a mean of 1.58%.



This slide shows the central reduction step. For each bed, we first run the pore-network model once and obtain its reference uptake curve. We then start the 1D model with a trial diffusivity D . The 1D solver produces a complete uptake curve, which we compare with the network curve. An optimization routine selects another D and repeats the transient 1D solve. The process continues until the NRMSE reaches its minimum. The minimizing value is the fitted D_{eff} for that bed.

In panel a, the solid lines are pore-network curves and the dashed lines are the optimized 1D curves. Panel b plots $\bar{q}_{1\text{D}} - \bar{q}_{\text{PNM}}$. Positive values indicate that the 1D model predicts too much uptake, and negative values indicate too little. NRMSE is the root-mean-square of this residual over time, normalized by the final network loading. It summarizes the mismatch of the complete curves, not only their final points. Across the twenty beds, the mean NRMSE is 1.58 percent, so one optimized coefficient reproduces each detailed network response closely.

The closure maps a cheap descriptor to the fitted coefficient

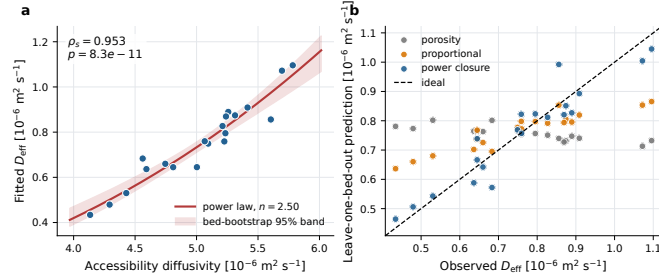
Closures compared

$$D_{\text{eff}} = a_0 + a_1 \varepsilon_b,$$

$$D_{\text{eff}} = \alpha D_{\text{access}},$$

$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

Complete beds are left out during fitting and then predicted.



(a) Selected power-law fit. ρ_s is the Spearman rank correlation; p tests the absence of a monotonic association.

(b) Held-out predictions from all three models; the proportional model is the orange series.

The positive power law gives LOOCV $R^2 = 0.890$; porosity gives $R^2 = -0.234$.

After the previous optimization, every bed has one fitted D_{eff} and one independently calculated D_{access} . We now develop a closure, meaning an equation that predicts the reduced-model coefficient from a quantity available from the packing.

We compare three equations. The first predicts D_{eff} from porosity. The second assumes direct proportionality to D_{access} . The third is a positive dimensionless power law. Panel a shows the selected power-law relationship. The Spearman coefficient ρ_s measures monotonic rank association, while the p value tests the hypothesis of no monotonic association. Panel b contains held-out predictions from all three models, including the proportional model shown in orange.

We use leave-one-bed-out cross-validation. Each bed is predicted by a closure fitted only to the other nineteen beds. The power law gives $R^2 = 0.890$, whereas the porosity model gives a negative R^2 and has no useful predictive ability.

Why not predict uptake directly from G_{access} ?

Direct endpoint relation

$$G_{\text{access}} \longrightarrow \bar{q}(5000 \text{ s})$$

- strong correlation: $\rho_s = 0.968$
- ranks beds or predicts one endpoint
- valid at the selected time and conditions

Transport closure

$$G_{\text{access}} \longrightarrow D_{\text{access}} \longrightarrow D_{\text{eff}} \longrightarrow \bar{q}(t)$$

- generates the complete uptake curve
- retains conservative 1D transport
- gives axial $c(z, t)$ and $q(z, t)$ profiles

If only uptake at 5000 s is needed, G_{access} is sufficient; D_{eff} is needed for a reusable transient model.

Why not predict uptake directly from G_{access} ?

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Direct endpoint relation	Transport closure
$G_{\text{access}} \longrightarrow \bar{q}(5000 \text{ s})$	$G_{\text{access}} \longrightarrow D_{\text{access}} \longrightarrow D_{\text{eff}} \longrightarrow \bar{q}(t)$
• strong correlation: $\rho_s = 0.968$ • ranks beds or predicts one endpoint • valid at the selected time and conditions	• generates the complete uptake curve • retains conservative 1D transport • gives axial $c(z, t)$ and $q(z, t)$ profiles
If only uptake at 5000 s is needed, G_{access} is sufficient; D_{eff} is needed for a reusable transient model.	

Accessibility conductance correlates very strongly with uptake at 5000 seconds. If our only task were to rank beds or estimate this one endpoint under the same conditions, we could use G_{access} directly.

Our intended reduced model asks for more. We want the complete uptake curve and the axial concentration and loading profiles. The 1D conservation equations require a transport coefficient, which is D_{eff} . Furthermore, D_{access} should not simply be inserted as D_{eff} because the strict proportional model performs substantially worse than the fitted power-law closure. The sequence is therefore: calculate D_{access} cheaply, predict D_{eff} through the closure, and use D_{eff} in the transient 1D model.

Why use an interior plane?

The accessibility calculation connects the inlet to an interior plane at a selected depth. It includes every connected path and every bottleneck between these two boundaries. Therefore, a bottleneck in the second layer or any deeper layer below the plane directly reduces G_{access} . We use an interior plane because the descriptor targets access from the inlet into the region sampled during the earlier part of adsorption. A plane at the bed top would instead measure steady conductance through the complete bed. It would include upper restrictions that affect the response mainly later in the transient. This does not mean that upper bottlenecks are physically unimportant. It means that whole-bed conductance answers a different question from inlet accessibility. The appropriate depth must therefore be checked rather than assumed.

Why use an interior plane?



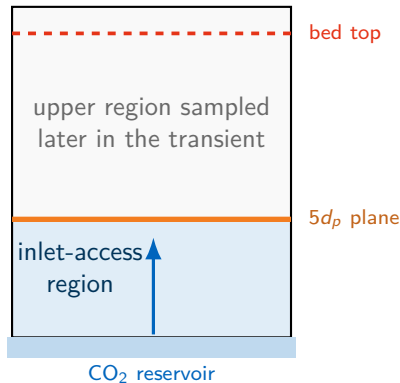
A plane at $5d_p$ includes

- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

The bed top answers a different question

- measure conductance through the entire bed
- include upper bottlenecks not sampled early

Why use an interior plane?



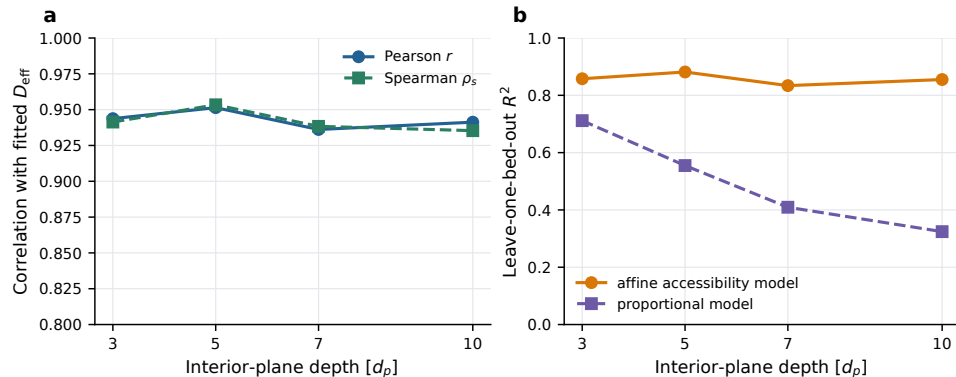
A plane at $5d_p$ includes

- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

The bed top answers a different question

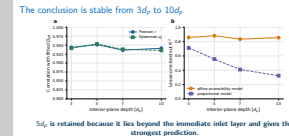
- measure conductance through the entire bed
- include upper bottlenecks not sampled early

The conclusion is stable from $3d_p$ to $10d_p$



$5d_p$ is retained because it lies beyond the immediate inlet layer and gives the strongest prediction.

The conclusion is stable from $3d_p$ to $10d_p$



We repeated the accessibility calculation using planes at three, five, seven and ten particle diameters from the inlet. The left panel shows that both Pearson and Spearman correlations with fitted D_{eff} remain high at all four depths. The right panel shows the corresponding leave-one-bed-out prediction scores.

The result is therefore not created by one arbitrary plane position. We retain five particle diameters because this plane lies beyond the immediate inlet layer, includes second-layer and deeper bottlenecks, and gives the strongest prediction in this data set. The proportional closure is weaker than the affine accessibility model at every tested depth, which also supports using a fitted nonlinear closure rather than identifying D_{access} directly with D_{eff} .

What the present evidence establishes

Supported

- porosity alone misses important packing variability
- inlet-connected topology controls transient utilization
- one D_{eff} reproduces each detailed curve accurately
- D_{access} provides an interpretable predictive closure

Not yet claimed

- industrial breakthrough performance
- effects of flow, heat or water competition
- transfer to other bead sizes or D_{tube}/d_p
- a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

2026-09-11

What the present evidence establishes

The present evidence supports four conclusions. Bulk porosity misses important differences among random packings. Inlet-connected topology explains a large part of the resulting variation. One optimized D_{eff} allows the 1D model to reproduce each detailed uptake curve accurately. Finally, D_{access} provides a cheap and interpretable input for predicting that coefficient. The scope must remain clear. We have not modelled an industrial breakthrough column with imposed flow. We have not included heat effects, water or competitive adsorption. We also have not demonstrated that the fitted power law transfers to other bead sizes or tube-to-particle ratios. The result is a validated structure-to-model closure for this controlled, diffusion-driven adsorption problem, rather than a universal adsorption correlation.

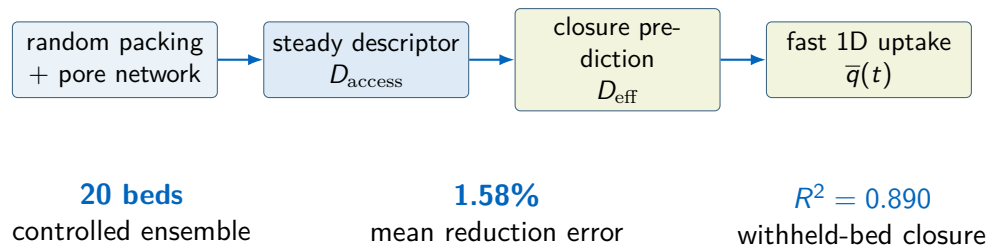
What the present evidence establishes

Supported	Not yet claimed
• porosity alone misses important packing variability • inlet-connected topology controls transient utilization • one D_{eff} reproduces each detailed curve accurately • D_{access} provides an interpretable predictive closure	• industrial breakthrough performance • effects of flow, heat or water competition • transfer to other bead sizes or D_{tube}/d_p • a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



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The main message is that equal void volume does not imply equal transport. How the void space connects to the inlet matters. Starting from a random packing, we extract the pore network and calculate the steady accessibility descriptor. The fitted closure converts this descriptor into D_{eff} , and the conservative 1D model uses that coefficient to predict the transient uptake.

Across twenty controlled beds, porosity varied by only 0.55 percent while uptake varied much more strongly. The optimized 1D model reproduced the pore-network curves with a mean NRMSE of 1.58 percent. The accessibility power law achieved a leave-one-bed-out R^2 of 0.890. The practical outcome is therefore a fast transient model that retains a clear and measurable link to the underlying packing structure.